

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY CHANGA

B.Tech. (CL/EC/ME/CE/EE/IT) Sem - I Examination, 2011-12

ME - 101 Engineering Graphics

Date: 10/12/2011, Saturday

Time: 1.30p.m to 4.30 p.m

Maximum Marks: 70

Instructions: (i) Attempt all the questions.

(ii) Make suitable assumptions if require

SECTION 1

Q.1 Choose appropriate answer by rewriting the question. [5]

- (i) If the distance of a point P from fixed point is more than fixed line then curve formed by point P known as,
(a) Ellipse (b) hyperbola (c) parabola (d) eccentricity
- (ii) If top view length of line PQ is equal to the true length, then
(a) $\beta = \phi$ (b) $\beta \neq \phi$ (c) $\beta \leq \phi$ (d) $\beta \geq \phi$
- (iii) In isometric projection, isometric axis are _____ apart from each other
(a) 60° (b) 120° (c) 72° (d) 90°
- (iv) If the axis of pentagonal pyramid is at 72° to its base, then solid is called
(a) Regular (b) oblique (c) auxiliary (d) non of this
- (v) A circular plate is inclined to HP by 60° and VP by 30° , so its side view is,
(a) ellipse (b) semi circle (c) line (d) circle

OR

- Q.1 i) Construct a full size plain scale of 1 dm, and indicate 23mm and 0.99 dm on it. [3]**
- ii) Give the application of AutoCAD [2]**

Q. 2 (a) A Line AB, 120 mm long, is parallel to H.P. and inclined to V.P. by 50° . Point B is 10mm above H.P. and 40mm on in front of V.P. Point A is behind V.P. Draw the projection of line AB. [8]

Q. 2(b) Draw an ellipse passing through all the four corners A, B, C & D of a rhombus having diagonals AC=110mm and BD=70mm. Use "Arcs of circles" Method and find its eccentricity. [10]

OR

Q. 2(b) A string is unwound from a circle of 20 mm diameter. Draw the locus of string P for unwinding the string's one turn. String is kept tight during unwound. Draw tangent & normal to the curve at any point. [10]

Q. 3 Fig. 1 shows pictorial view of an object. Draw [12]
(1) Front view in the direction 'X', (2) Top view.

SECTION 2

- Q.4 (a)** An isosceles triangle plate of 50mm base and 75mm altitude appears as an equilateral triangle of 50mm in top view. Draw the projection of plate if its 50mm long edge is on the H.P. and inclined at 45° to V.P. [7]
- Q.4(b)** A pentagonal pyramid, base 25 mm side and axis 50 mm long has one of its faces in the V.P. and the edge of the base contained by that face makes an angle of 30° with the H.P. Draw its Projection. [10]
- OR**
- Q.4 (a)** Draw the projection of a regular hexagon of 25mm side, having one of its sides in the H.P. and inclined at 60° to the V.P., and its surface making an angle of 45° with the H.P. [7]
- Q.4(b)** A square pyramid of 50 mm side of base and 50 mm length of axis is resting on one of its triangular faces on the H.P. having a slant edge containing that face parallel to the V.P. Draw the projection of the pyramid. [10]
- Q.5** A cone, diameter of the base 60 mm and axis 70 mm long is resting on its base on the H.P. It is cut by an A.I.P. so that the true shape of the section is an isosceles triangle having 50 mm base. Draw the sectional top view, front view and true shape of the section. [8]
- OR**
- Q.5** A square prism, base 40 mm side, axis 80 mm long has its base on the H.P. and its faces equally inclined to the V.P. It is cut by a plane, perpendicular to the V.P., inclined at 60° to the H.P. and passing through a point on the axis, 55 mm above the H.P. Draw its front view, sectional top view, another top view of section on an AIP parallel to the section plane and development of the lateral surface of the remaining part prism. [8]
- Q.6** A frustum of a square pyramid has its base 80 mm side and top 40 mm side and height 50 mm. Draw isometric View of the frustum if its axis is parallel to H.P. and Perpendicular to V.P., edges of base are equally inclined to H.P. [10]

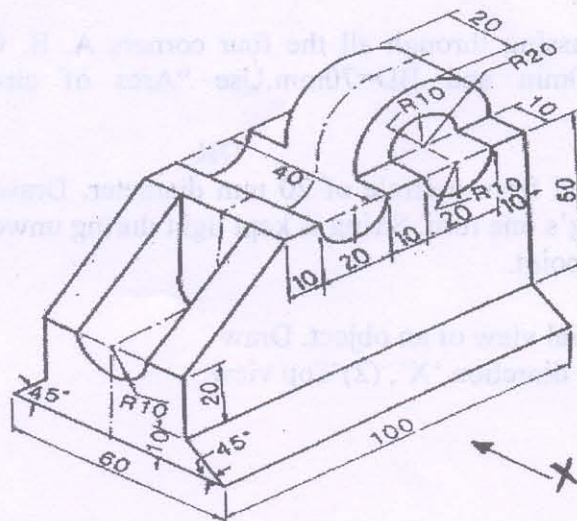


FIG.1